Data Portal - AI & Spec-Driven Approach

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Agenda

01 SDLC vs AI-DLC

02 Data Portal : AI-DLC

03 Before & After Metrics

04 Example - Release Notes

05 Example - QA Agent & Test Cases

06 Knows Risks & Challenges

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SDLC vs. AI-DLC

Standard SDLC (Manual)	AI-DLC (Using Spec-Driven Approach)
Requirements & Analysis Manual stakeholder interviews & documentation	**Intake & Spec Agent** Agent requirement analysis & spec generation
Architecture & Design Manual technical planning & diagrams	**Planner & Architect Agent** Automated codebase research & implementation plans
Code Development Human engineers writing line-by-line code	**Implementation Agent** AI writing production-grade code against specs
QA & Security Review Manual testing & periodic security audits	**Verification & Security Agents** Continuous automated tests
Deployment & Release Notes Manual deployment & provisioning of Release Notes	**Deployment & Release Notes** Automated provisioning of Release Notes

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Data Portal : AI-DLC

LEAD ENGINEER (Manager)

Routes work, validates gates, blocks bad code

Requirements Analyst (Intake)
Interviews users → Jira epic + Confluence page

Planner (Architecture)
Researches codebase → writes implementation plan

Implementer (Development)
Writes code per plan, verifies build passes

Test Writer (Quality)
Writes unit + E2E tests, fixes failures

Code Reviewer (Standards)
Enforces TypeScript, React, naming rules

Security Reviewer (Security)
Trivy & Top 10 checks before merge

Tester (Verification)
Final build + full test suite pass

Deployer (PR)
PR + health checks + Jira closure

Release Notes
Auto-generates release notes

More Agents...

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Data Portal - Metrics

Metric	Before (Traditional)	After (Spec-Driven)
Feature cycle time	14-20 days	**~3 days**
Rework cycles	5-8	**1-3**
Security bugs caught	~40%	**80 - 100%**
Standards violations	20-30%	**0% (gate-blocked)**
Test coverage	60-70%	**90% (mandatory)**
Onboarding time	Weeks (tribal knowledge)	**Hours (standards in repo)**
Deployment confidence	Medium (manual)	**High (automated + verified)**

Timeframe : 2 Sprints

Implementation for following features : Agentic Dashboard, CAS Engineering Excellence Dashboard

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Data Portal - Release Notes

The Workflow

1. Verify on master branch
2. Read git log since last release
3. Determine version bump
   - Patch: v2.2.0 → v2.2.1 (bug fix)
   - Minor: v2.2.0 → v2.3.0 (feature)
4. Categorize commits by area
5. Generate structured notes
6. Write local markdown file
7. Publish to Confluence

Trigger

Developer merges PR to master branch

The Release Notes Agent activates automatically to generate versioned documentation from git history.

Example Output :

CONFLUENCE PAGE

https://newrelic.atlassian.net/wiki/spaces/DATA/pages/5424513559/Release+v2.2.2

Zero manual effort. Every release documented. Every time.

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Data Portal - QA Agent

Pipeline Flow

- Feature Request
- Requirements → Spec
- Implementation
- **QA Agent E2E Tests**
- Code & Security Review
- Deploy to Production
- Release Notes Agent

E2E Verification

✓ Happy path tested

✓ Validation tested

✓ Edge cases tested

✓ No regressions

Result: Users experience exactly what was tested.

Release Evidence

✓ Traceable markdown

✓ Shareable Confluence

✓ Categorized by area

✓ Files changed listed

Output: e2e test confluence

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Known Risks & Challenges

Hallucinations
Agents can generate plausible but incorrect code

Context Drift
Long chains can lose earlier decisions

Over-Engineering
Agents may add unnecessary abstractions

Security Blind Spots
May reproduce insecure patterns from internet code

Test Coverage Gaps
Happy-path tests may mask failure modes

Runaway Actions
Risk of unintentional API calls or deletions

Review Burden Shifts: Reviewing agent output still requires engineering expertise

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References

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SDLC vs AIDLC

Comparing the traditional linear software lifecycle with the accelerated, AI-assisted development pipeline.

Standard SDLC

Requirement (Jira) — Analysis (Confluence) — Design (Confluence) — Planning (Jira) — Impl. (Code) — Testing (Code) — Deploy

Agentic AIDLC

AI	AI	AI	AI	AI	AI	
Requirement Intake Agent	**Analysis** Logic Agent	**Design** Arch. Agent	**Planning** Plan Agent	**Impl.** Coding Agent	**Testing** Test Agent	**Release** Auto-Notes

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The Feedback Loop using Specs

IMPLEMENTER

Builds & Learns

- Reads common-mistakes.md
- Avoids known issues
- Mistakes caught by reviewer

CODE REVIEWER

Validates Logic

- Marks BLOCKING issues
- Documents root cause
- Updates mistake repo

SECURITY REVIEWER

Audits Safety

- Flags vulnerabilities
- Categorizes (OWASP)
- Adds fix patterns

Mistake Log Grows

Collective intelligence scales

The Compounding Effect

Fewer issues reach review phase as the knowledge base expands. Every failure becomes a permanent preventative measure for the entire fleet.

Review Cycles Decrease Over Time

Month 1: 8 blocking issues / feature

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Spec-Driven Development (SDD) Flow

Rework Loop: Fails gate → Auto-correction

HUMAN

1. Specification

Human-defined requirements & logic and _state.md spec file created

AGENT

2. Implementation

Agent writes code against spec and updates _state.md

AGENT

3. Verification

Automated tests & gate checks

HUMAN

4. Approval

Final human review & merge

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The Memory Stack — specs in action

Agents don't just read one document — they read a layered knowledge(spec) stack

LAYER 4

common-mistakes.md — "What has gone wrong before?"

• Specific defects with root cause | Grows with every caught mistake | Read by EVERY agent

LAYER 3

quality-checks.md — "What must be verified?"

• TypeScript/React checklists | Security (OWASP) | Gate criteria for Reviewers

LAYER 2

coding-standards.md — "How should code be written?"

• Naming conventions | Component structure | Error handling patterns

LAYER 1

tech-stack.md — "What tools and constraints exist?"

• Frameworks | Key paths | Environment variables | Foundational for ALL agents

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Built-In Quality & Safety Strategies

Risk	Mitigation
Incorrect code	Dedicated code-reviewer agent + human PR review
Context drift	_state.md read at every phase handoff
Security gaps	Security-reviewer agent runs OWASP Top 10 on every PR
Runaway actions	4 human approval gates; no auto-push, no auto-merge
Test gaps	Negative tests + edge cases required; bun run build is the gate
Secrets in code	Pre-commit hook scans for API keys and tokens
Rubber-stamping	Agent PRs treated identically to human PRs — line-by-line review required

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AI Does the Work. Humans Stay in Control

1. **Requirements approval** — before planning begins
2. **Architecture plan approval** — before any code is written
3. **Security review sign-off** — any critical finding blocks the pipeline
4. **Final PR merge** — human reviews and approves
5. **Production deployments** — human initiates and monitors, agent creates release notes

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Where We Go From Here

Current Status & Expansion

- Pipeline live for Data Portal development
- Ready to scale to Data engineering and Support

Optimization & Learning

- Velocity metrics collection underway
- Investment in memory (common-mistakes.md)

Strategic Ask of Leadership

- **Support adoption:** Drive agentic workflows across Data portal in DTM
- **Allocate time:** Empower engineers to supervise and improve agent quality
- **Endorse the model:** Establish this pipeline as the template for other teams

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Thank You

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